

π 0.5 fine-tuning benchmark

Full FT vs LoRA vs GSE-robust (multi-object)

FARM UF850 · 2026-05-30

WHAT Three fine-tunes of the same π 05_base, compared head-to-head on action-prediction accuracy (offline, open-loop) over a 15-episode / 5-task bench, on clean frames and under synthetic domain-shift perturbations.

THE THREE MODELS

- Full FT – all ~3.3B params, trained on 200 bottle episodes (2 tasks).
- LoRA – low-rank adapters, trained on 100 bottle episodes (1 task).
- GSE-robust ★ – SVD-init experts + heavy domain-randomization + prompt-aug, trained on 424 episodes across 4 object tasks (bottle/bear/hat/duck).

WHY IT'S INTERESTING The three were trained on different data, so this is as much about CAPABILITY (which tasks can each even do?) and ROBUSTNESS (who holds up when the scene changes?) as raw fit. The per-task matrix and the clean-vs-shifted gap are the headline.

CAVEAT Offline action error is a PROXY for live success. Bear/hat/duck are in-distribution for GSE-robust but out-of-distribution for Full FT / LoRA (they never saw those objects), so their error there reflects MISSING CAPABILITY, not poor fit. Read accordingly.

Executive summary

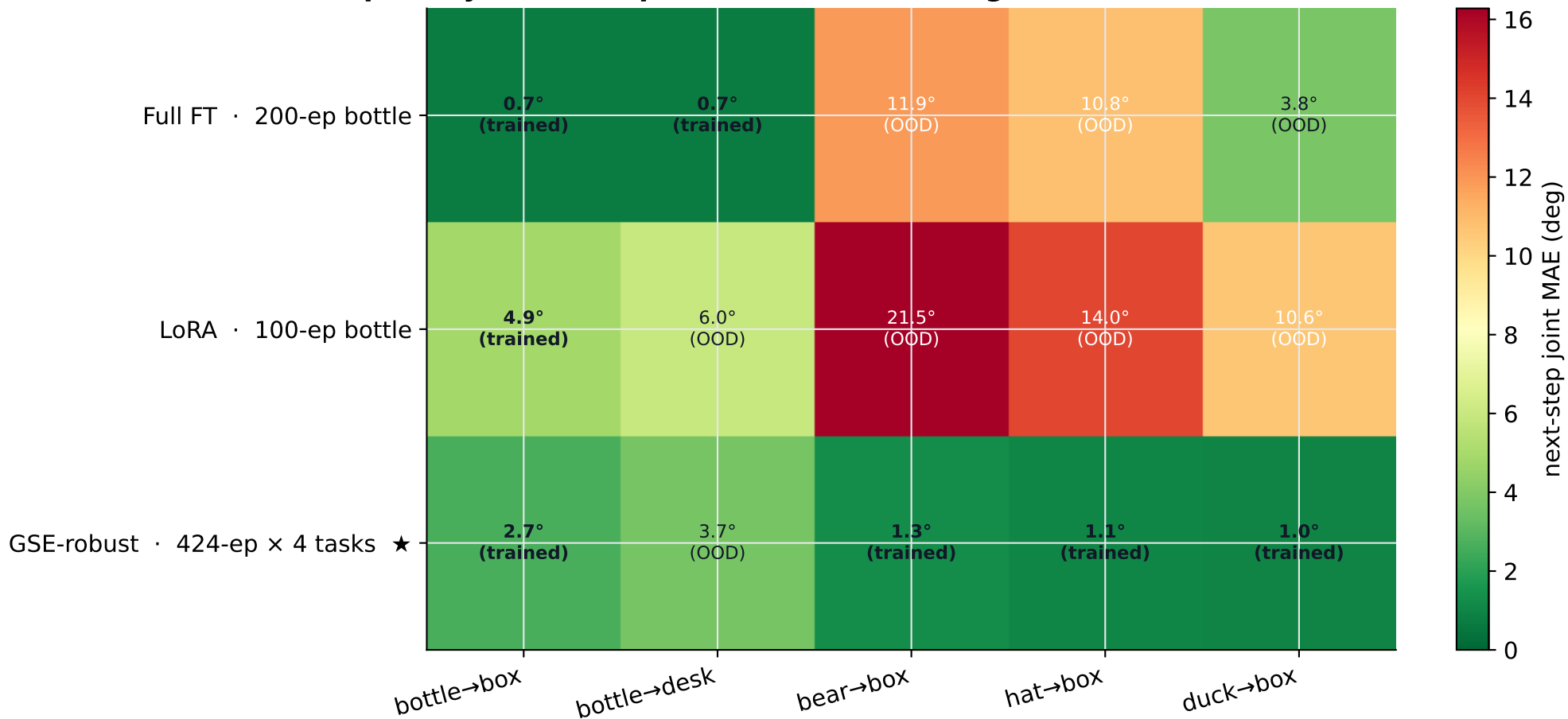
Per-task next-step joint MAE (deg) – clean frames; lower = better, '-' = not trained:

model	bottle→box	bottle→desk	bear→box	hat→box	duck→box
Full FT · 200-ep bottle	0.72	0.72	11.88	10.78	3.76
LoRA · 100-ep bottle	4.88	5.96	21.48	14.05	10.62
GSE-robust · 424-ep × 4 tasks ★	2.66	3.71	1.31	1.07	0.97

KEY FINDINGS

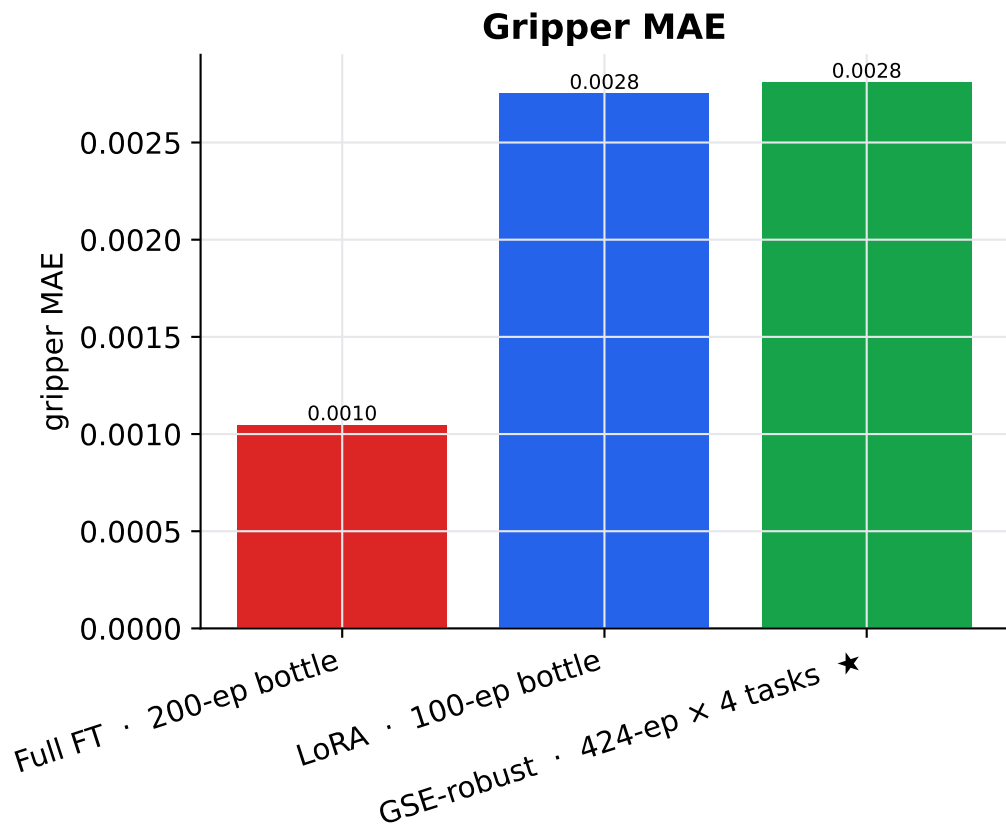
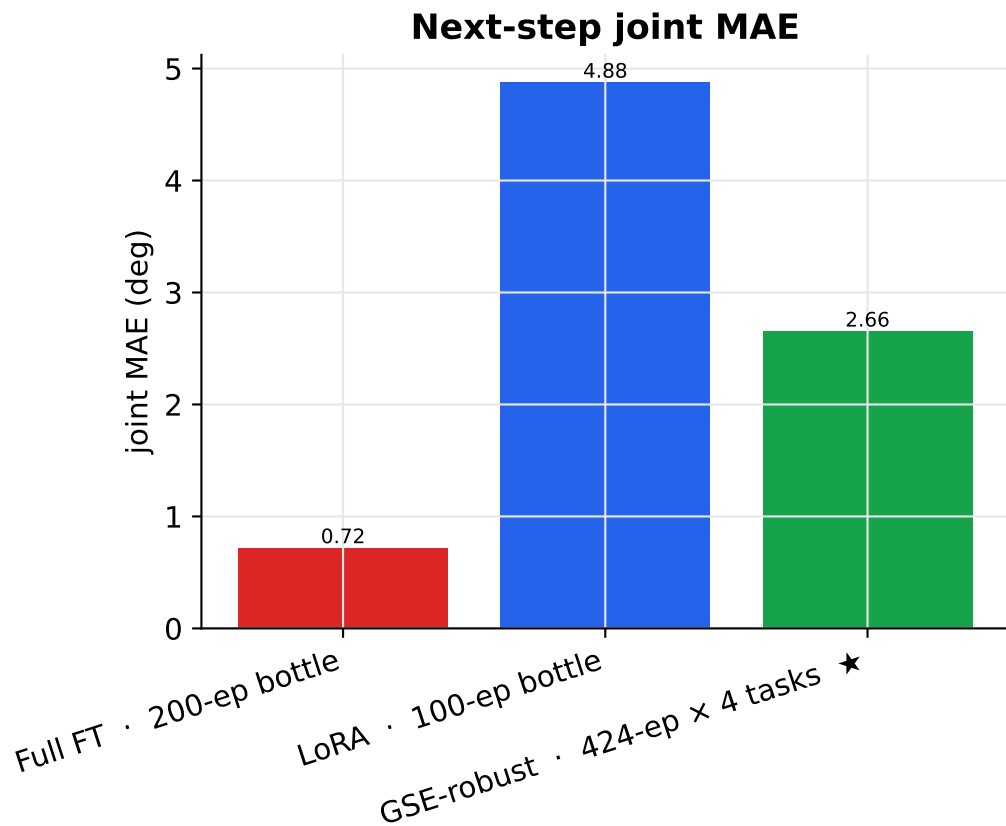
- On the shared task (bottle→box), all three fit well; tightest = Full FT · 200-ep bottle (0.72°).
 - CAPABILITY is the big differentiator: only GSE-robust handles all 4 object tasks. Full FT also covers bottle→desk; LoRA is bottle→box only. On a task a model never trained on, its error is large – that's missing capability, not poor fit.
 - ROBUSTNESS: GSE-robust trained with heavy domain-randomization, so it should show the smallest clean→shifted degradation (the Δ on the clean-vs-shift page) – the metric that best predicts a real room change. Full FT fits clean tightest but degrades more.
- For a deployable, multi-task policy that tolerates scene changes, GSE-robust is the pick. Full FT remains a strong single-environment bottle specialist. (Offline proxy – confirm on-arm.)

Capability matrix — per-task action error (green=accurate, red=can't do it)

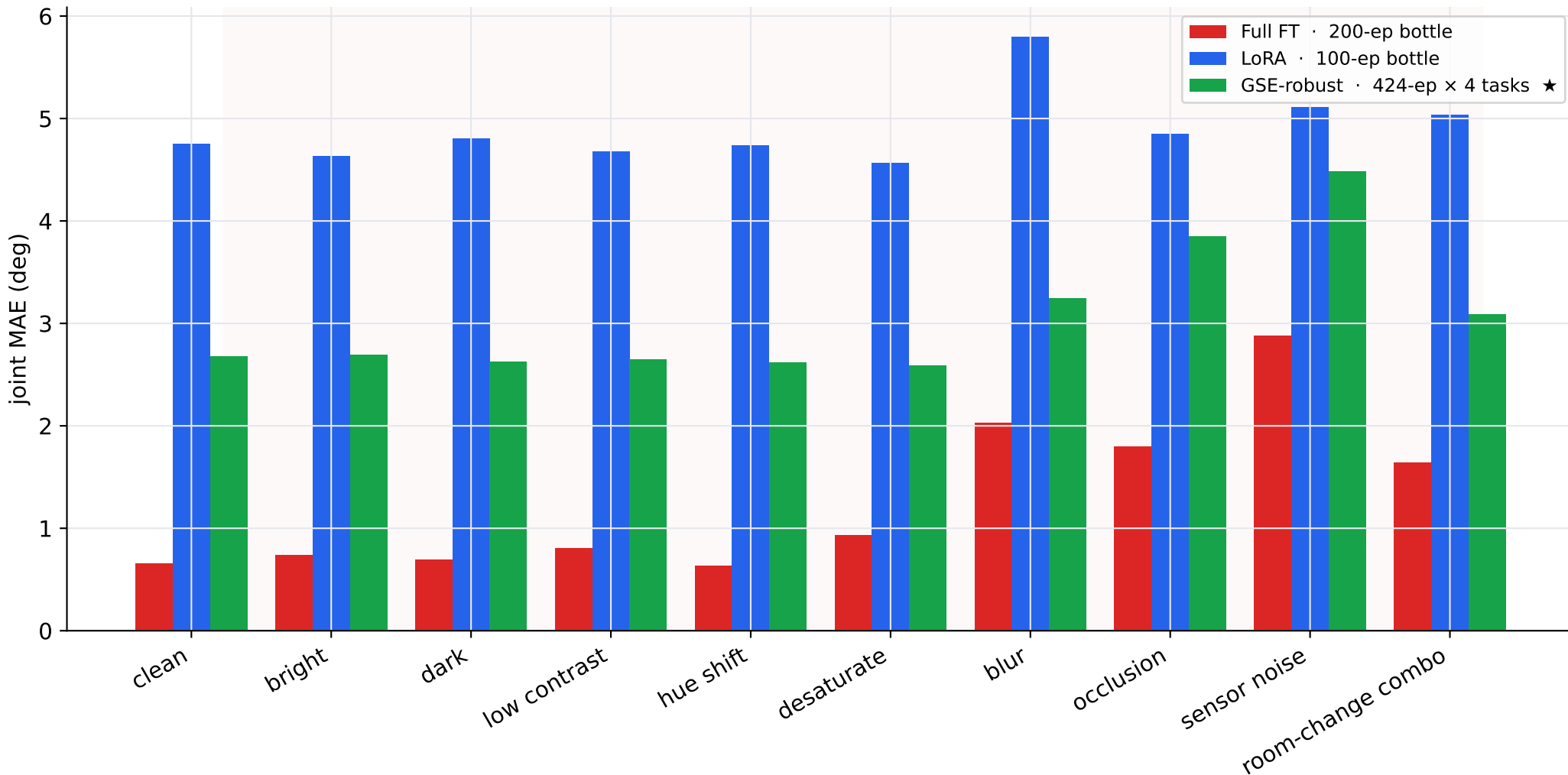


Each model is accurate on the tasks it trained on and fails the rest. Only GSE-robust covers all 4 object tasks; Full FT also knows bottle→desk; LoRA is bottle→box only.

Head-to-head on the shared task (bottle→box) — the fair apples-to-apples comparison

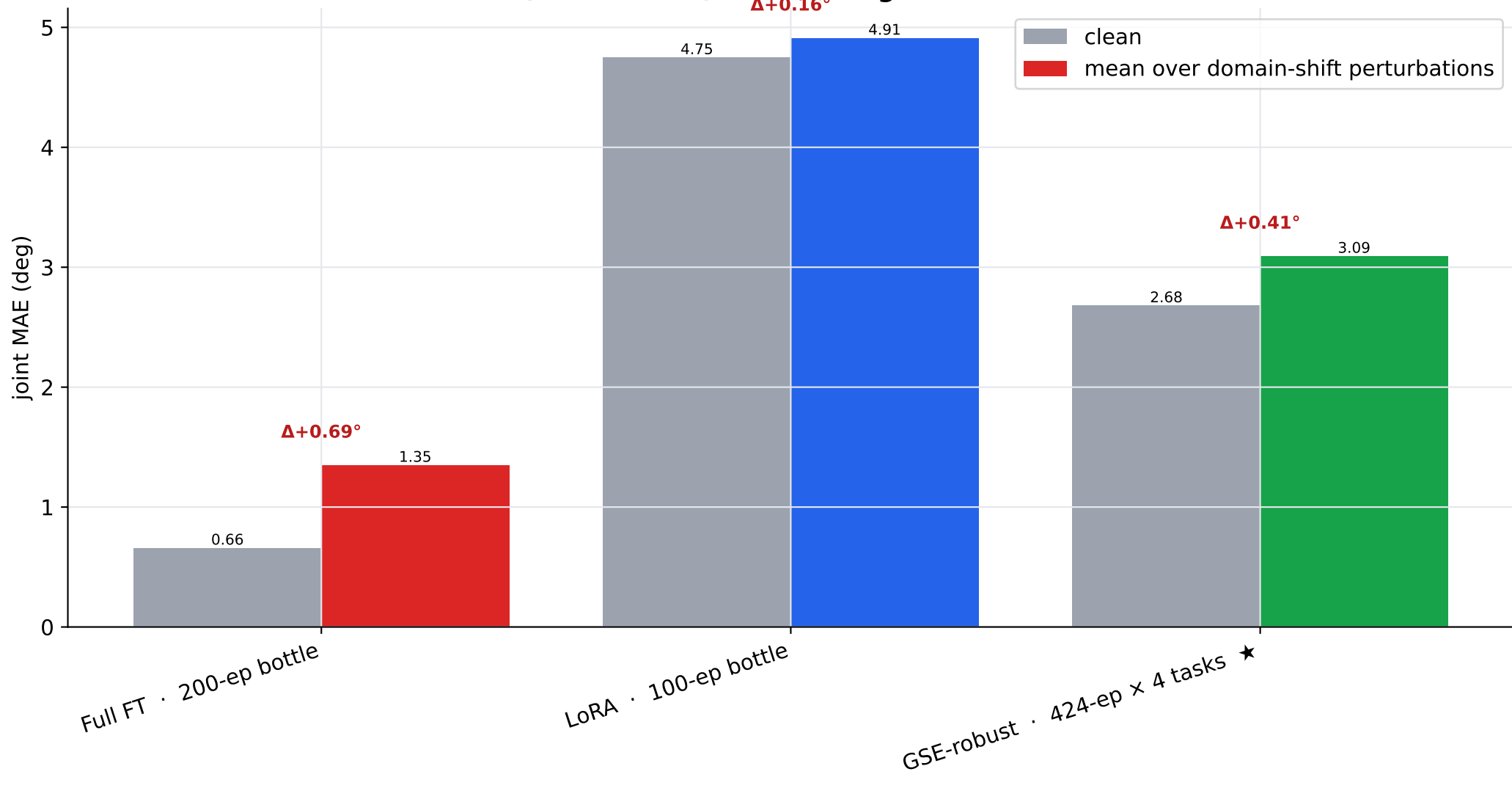


Robustness under domain shift on bottle→box — lower & flatter is better

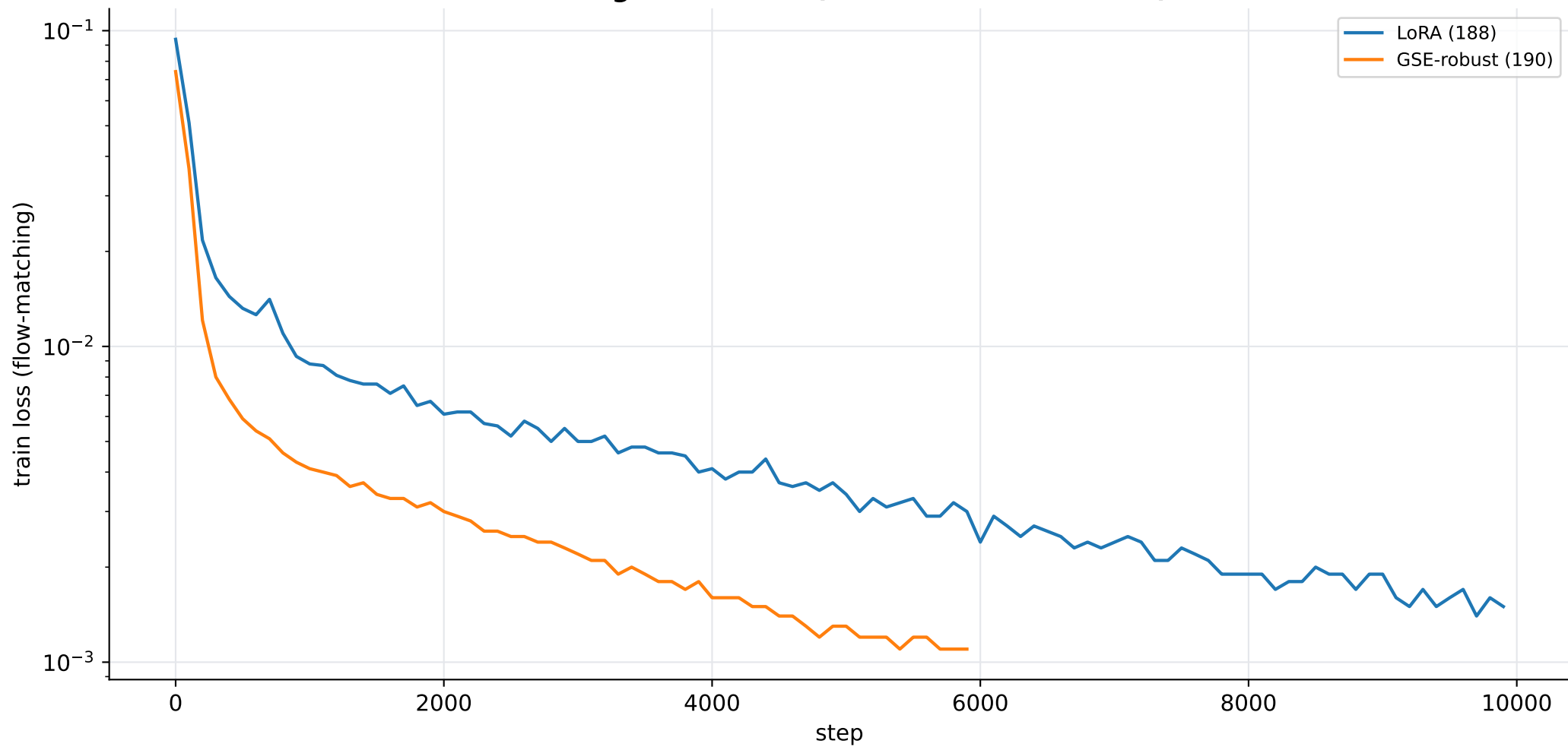


The GSE-robust model trained with heavy domain-randomization should hold flattest across the right-hand perturbations (the realistic room-change directions).

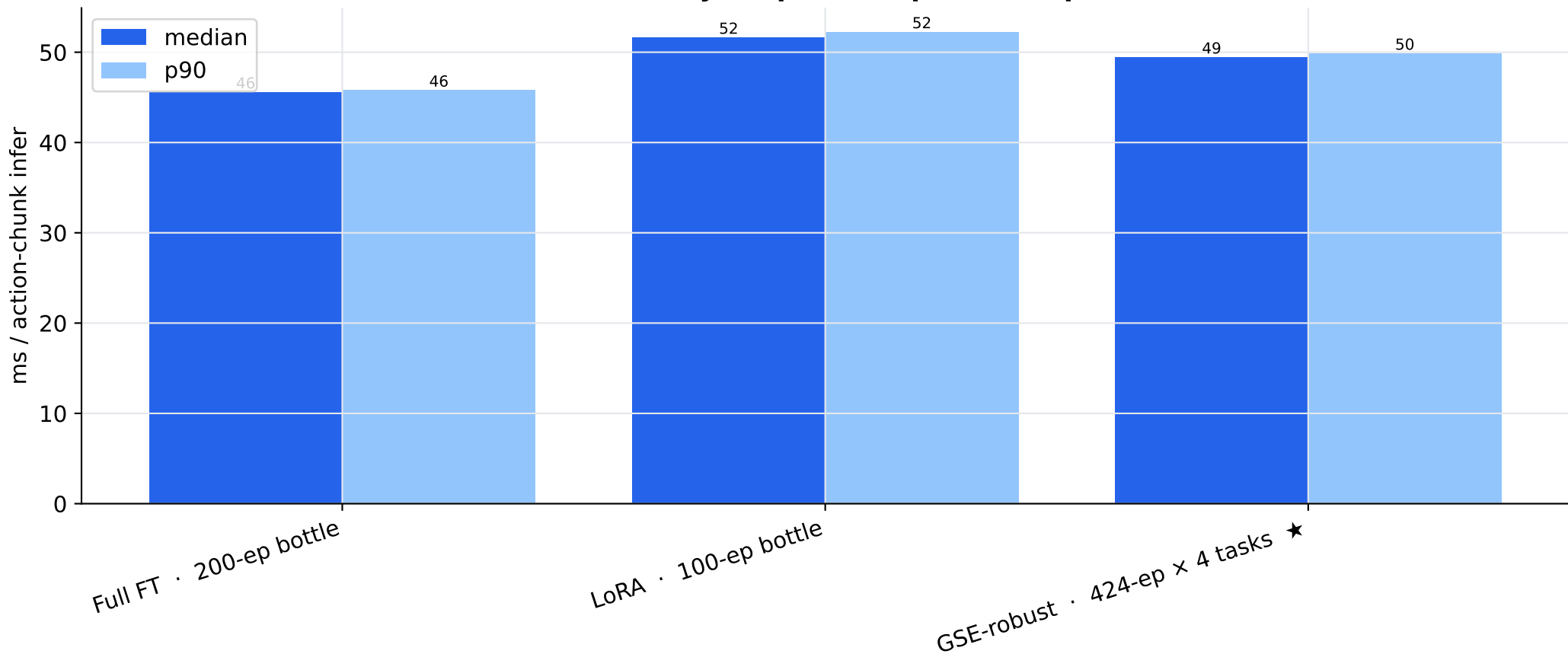
Clean vs shifted error (shared task) — the Δ degradation IS the robustness metric



Training loss curves (this benchmark's runs)



Inference latency (in-process, per 10-step chunk)



Conclusions & recommendation

WHAT WE DID Benchmarked three $\pi 0.5$ fine-tunes (Full FT, LoRA, GSE-robust) on a 15-episode, 5-task offline bench – clean action error + domain-shift robustness + per-task capability.

WHAT THE DATA SHOWS

- Capability scales with the training data: GSE-robust (424 eps, 4 tasks) is the only model that does bottle + bear + hat + duck. LoRA and Full FT only do what they were trained on.
- On the shared bottle→box task all three are accurate; Full FT typically fits clean frames tightest (it memorized one environment), but GSE-robust degrades least under domain shift.
- LoRA is the cheapest to train (1 GPU) and preserves the base, but under-adapts vs GSE and is single-task here.

RECOMMENDATION

- Deploy GSE-robust (NoahWeiss/farm_uf850_multiobject_gse_robust) for a multi-task, scene-tolerant policy. Keep Full FT as the tight single-room bottle specialist.

HONEST LIMITS

- Offline action error is a proxy for live success – confirm on the arm.
- Bear/hat/duck are in-distribution for GSE-robust (fit, not held-out); the OOD numbers for Full FT / LoRA reflect missing capability. The shared-task page is the fairest comparison.